

DPP NO.-10

Choose the correct alternative (only one is correct):

- Q.1 There are 100 different books in a shelf. Number of ways in which 3 books can be selected so that no two of which are neighbours is
(A) $^{100}C_3 - 98$ (B) $^{97}C_3$ (C) $^{96}C_3$ (D) $^{98}C_3$
- Q.2 Two classrooms A and B having capacity of 25 and $(n-25)$ seats respectively. A_n denotes the number of possible seating arrangements of room 'A', when 'n' students are to be seated in these rooms, starting from room 'A' which is to be filled up full to its capacity.
If $A_n - A_{n-1} = 25! (^{49}C_{25})$ then 'n' equals
(A) 50 (B) 48 (C) 49 (D) 51
- Q.3 The sum of all numbers greater than 1000 formed by using digits 1, 3, 5, 7 no digit being repeated in any number is :
(A) 72215 (B) 83911 (C) 106656 (D) 114712
- Q.4 Number of positive integral solutions satisfying the equation $(x_1 + x_2 + x_3)(y_1 + y_2) = 77$, is
(A) 150 (B) 270 (C) 420 (D) 1024
- Q.5 Distinct 3 digit numbers are formed using only the digits 1, 2, 3 and 4 with each digit used at most once in each number thus formed. The sum of all possible numbers so formed is
(A) 6660 (B) 3330 (C) 2220 (D) none
- Q.6 The streets of a city are arranged like the lines of a chess board . There are m streets running North to South & 'n' streets running East to West . The number of ways in which a man can travel from NW to SE corner going the shortest possible distance is :
(A) $\sqrt{m^2 + n^2}$ (B) $\sqrt{(m-1)^2 + (n-1)^2}$ (C) $\frac{(m+n)!}{m! \cdot n!}$ (D) $\frac{(m+n-2)!}{(m-1)! \cdot (n-1)!}$
- Q.7 An ice cream parlour has ice creams in eight different varieties . Number of ways of choosing 3 ice creams taking atleast two ice creams of the same variety, is :
(A) 56 (B) 64 (C) 100 (D) none
(Assume that ice creams of the same variety are identical & available in unlimited supply)
- Q.8 There are 12 books on Algebra and Calculus in our library , the books of the same subject being different. If the number of selections each of which consists of 3 books on each topic is greatest then the number of books of Algebra and Calculus in the library are respectively:
(A) 3 and 9 (B) 4 and 8 (C) 5 and 7 (D) 6 and 6
- Q.9 The sum of all the numbers formed from the digits 1, 3, 5, 7, 9 which are smaller than 10,000 if repetition of digits is not allowed, is
(A) (28011)S (B) (28041)S (C) (28121)S (D) (29152)S
where $S = (1+3+5+7+9)$

Choose the correct alternatives (More than one are correct):

- Q.10 The combinatorial coefficient $C(n, r)$ is equal to
(A) number of possible subsets of r members from a set of n distinct members.
(B) number of possible binary messages of length n with exactly r 1's.
(C) number of non decreasing 2-D paths from the lattice point (0, 0) to (r, n).
(D) number of ways of selecting r things out of n different things when a particular thing is always included plus the number of ways of selecting 'r' things out of n, when a particular thing is always excluded.

- Q.11 Identify the correct statement(s).
 (A) Number of naughts standing at the end of 125 is 30 .
 (B) A telegraph has 10 arms and each arm is capable of 9 distinct positions excluding the position of rest. The number of signals that can be transmitted is $10^{10} - 1$.
 (C) Number of numbers greater than 4 lacs which can be formed by using only the digits 0, 2, 2, 4, 4 and 5 is 90.
 (D) In a table tennis tournament, every player plays with every other player. If the number of games played is 5050 then the number of players in the tournament is 100.
- Q.12 There are 10 questions, each question is either True or False. Number of different sequences of incorrect answers is also equal to
 (A) Number of ways in which a normal coin tossed 10 times would fall in a definite order if both Heads and Tails are present.
 (B) Number of ways in which a multiple choice question containing 10 alternatives with one or more than one correct alternatives, can be answered.
 (C) Number of ways in which it is possible to draw a sum of money with 10 coins of different denominations taken some or all at a time.
 (D) Number of different selections of 10 indistinguishable things taken some or all at a time.
- Q.13 The continued product, $2 \cdot 6 \cdot 10 \cdot 14 \dots$ to n factors is equal to
 (A) ${}^{2n}C_n$ (B) ${}^{2n}P_n$ (C) $(n+1)(n+2)(n+3) \dots (n+n)$ (D) none
- Q.14 The Number of ways in which five different books to be distributed among 3 persons so that each person gets at least one book, is equal to the number of ways in which
 (A) 5 persons are allotted 3 different residential flats so that and each person is allotted at most one flat and no two persons are allotted the same flat.
 (B) number of parallelograms (some of which may be overlapping) formed by one set of 6 parallel lines and other set of 5 parallel lines that goes in other direction.
 (C) 5 different toys are to be distributed among 3 children, so that each child gets at least one toy.
 (D) 3 mathematics professors are assigned five different lectures to be delivered, so that each professor gets at least one lecturer.
- Q.15 The combinatorial coefficient ${}^{n-1}C_p$ denotes
 (A) the number of ways in which n things of which p are alike and rest different can be arranged in a circle.
 (B) the number of ways in which p different things can be selected out of n different thing if a particular thing is always excluded.
 (C) number of ways in which n alike balls can be distributed in p different boxes so that no box remains empty and each box can hold any number of balls.
 (D) the number of ways in which $(n-2)$ white balls and p black balls can be arranged in a line if black balls are separated, balls are all alike except for the colour.
- Q.16 The maximum number of permutations of $2n$ letters in which there are only a's & b's, taken all at a time is given by :
 (A) ${}^{2n}C_n$ (B) $\frac{2}{1} \cdot \frac{6}{2} \cdot \frac{10}{3} \dots \frac{4n-6}{n-1} \cdot \frac{4n-2}{n}$
 (C) $\frac{n+1}{1} \cdot \frac{n+2}{2} \cdot \frac{n+3}{3} \cdot \frac{n+4}{4} \dots \frac{2n-1}{n-1} \cdot \frac{2n}{n}$ (D) $\frac{2^n \cdot [1 \cdot 3 \cdot 5 \dots (2n-3)(2n-1)]}{n!}$

- Q.17 Number of ways in which 3 numbers in A.P. can be selected from $1, 2, 3, \dots, n$ is :
- (A) $\left(\frac{n-1}{2}\right)^2$ if n is even (B) $\frac{n(n-2)}{4}$ if n is odd
- (C) $\frac{(n-1)^2}{4}$ if n is odd (D) $\frac{n(n-2)}{4}$ if n is even
- Q.18 If $P(n, n)$ denotes the number of permutations of n different things taken all at a time then $P(n, n)$ is also identical to
- (A) $r! \cdot P(n, n-r)$ (B) $(n-r) \cdot P(n, r)$ (C) $n \cdot P(n-1, n-1)$ (D) $P(n, n-1)$ where $0 \leq r \leq n$
- Q.19 Which of the following statements are correct?
- (A) Number of words that can be formed with 6 only of the letters of the word "CENTRIFUGAL" if each word must contain all the vowels is $3 \cdot 7!$
- (B) There are 15 balls of which some are white and the rest black. If the number of ways in which the balls can be arranged in a row, is maximum then the number of white balls must be equal to 7 or 8. Assume balls of the same colour to be alike.
- (C) There are 12 things, 4 alike of one kind, 5 alike and of another kind and the rest are all different. The total number of combinations is 240.
- (D) Number of selections that can be made of 6 letters from the word "COMMITTEE" is 35.

MATCH THE COLUMN

- Q.20
- | Column I | Column II |
|---|------------------|
| (A) Number of increasing permutations of m symbols are there from the n set numbers $\{a_1, a_2, \dots, a_n\}$ where the order among the numbers is given by $a_1 < a_2 < a_3 < \dots < a_{n-1} < a_n$ is | (P) n^m |
| (B) There are m men and n monkeys. Number of ways in which every monkey has a master, if a man can have any number of monkeys | (Q) ${}^m C_n$ |
| (C) Number of ways in which n red balls and $(m-1)$ green balls can be arranged in a line, so that no two red balls are together, is (balls of the same colour are alike) | (R) ${}^n C_m$ |
| (D) Number of ways in which ' m ' different toys can be distributed in ' n ' children if every child may receive any number of toys, is | (S) m^n |
- Q.21
- | Column-I | Column-II |
|--|------------------|
| (A) Four different movies are running in a town. Ten students go to watch these four movies. The number of ways in which every movie is watched by atleast one student, is (Assume each way differs only by number of students watching a movie) | (P) 11 |
| (B) Consider 8 vertices of a regular octagon and its centre. If T denotes the number of triangles and S denotes the number of straight lines that can be formed with these 9 points then the value of $(T-S)$ equals | (Q) 36 |
| (C) In an examination, 5 children were found to have their mobiles in their pocket. The Invigilator fired them and took their mobiles in his possession. Towards the end of the test, Invigilator randomly returned their mobiles. The number of ways in which at most two children did not get their own mobiles is | (R) 52 |
| (D) The product of the digits of 3214 is 24. The number of 4 digit natural numbers such that the product of their digits is 12, is | (S) 60 |
| (E) The number of ways in which a mixed double tennis game can be arranged from amongst 5 married couple if no husband & wife plays in the same game, is | (T) 84 |

SUBJECTIVE :

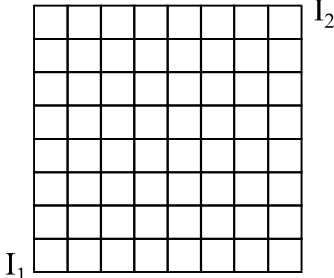
- Q.22 A committee of 10 members is to be formed with members chosen from the faculties of Arts, Economics, Education, Engineering, Medicine and Science. Number of possible ways in which the faculties representation be distributed on this committee, is _____.
(Assume every department contains more than 10 members).
- Q.23 How many ways are there to seat n married couples ($n \geq 3$) around a table such that men and women alternate and each women is not adjacent to her husband.
- Q.24 10 identical ball are distributed in 5 different boxes kept in a row and labled A, B, C, D and E. Find the number of ways in which the ball can be distributed in the boxes if no two adjacent boxes remain empty.
- Q.25 The number of non negative integral solution of the inequation $x + y + z + w \leq 7$ is ____.
- Q.26 On the normal chess board as shown, I_1 & I_2 are two insects which starts moving towards each other. Each insect moving with the same constant speed. Insect I_1 can move only to the right or upward along the lines while the insect I_2 can move only to the left or downward along the lines of the chess board. Prove that the total number of ways the two insects can meet at same point during their trip is equal to
- $$\left(\frac{9}{8}\right) \left(\frac{10}{7}\right) \left(\frac{11}{6}\right) \left(\frac{12}{5}\right) \left(\frac{13}{4}\right) \left(\frac{14}{3}\right) \left(\frac{15}{2}\right) \left(\frac{16}{1}\right)$$

OR

$$2^8 \left(\frac{1}{1}\right) \left(\frac{3}{2}\right) \left(\frac{5}{3}\right) \left(\frac{7}{4}\right) \left(\frac{9}{5}\right) \left(\frac{11}{6}\right) \left(\frac{13}{7}\right) \left(\frac{15}{8}\right)$$

OR

$$\left(\frac{2}{1}\right) \left(\frac{6}{2}\right) \left(\frac{10}{3}\right) \left(\frac{14}{4}\right) \left(\frac{18}{5}\right) \left(\frac{22}{6}\right) \left(\frac{26}{7}\right) \left(\frac{30}{8}\right)$$


- Q.27 How many numbers gretater than 1000 can be formed from the digits 112340 taken 4 at a time.
- Q.28 Tom has 15 ping-pong balls each uniquely numbered from 1 to 15. He also has a red box, a blue box, and a green box.
- (a) How many ways can Tom place the 15 distinct balls into the three boxes so that no box is empty?
- (b) Suppose now that Tom has placed 5 ping-pong balls in each box. How many ways can he choose 5 balls from the three boxes so that he chooses at least one from each box?
- Q.29 Find the number of ways in which 12 identical coins can be distributed in 6 different purses, if not more than 3 & not less than 1 coin goes in each purse.
- Q.30 A drawer is fitted with n compartments and each compartment contains n counter, no two of which marked alike. Number of combinations which can be made with these counters if no two out of the same compartment enter into any combination, is _____.
- Q.31 In how many ways it is possible to select six letters, including at least one vowel from the letters of the word "F L A B E L L I F O R M". (It is a picnic spot in U.S.A.)